Boundary-Energy Certificates for Uniform Real Lipschitz Surfaces

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Abstract

We analyze the conditional-covariance and boundary-sampling interfaces in the assigned manuscript on uniform real Lipschitz heights on the triangular lattice. The model is Lebesgue measure on a hard-constraint polytope, not an integer height model or a uniformly convex gradient field. Retaining the source's continuum-thickness and Beurling inputs, we apply killing estimates at both endpoints of the Dirichlet Green kernel. For a positive measure with a uniform one-half Frostman bound supported within distance ε of the observed set, this gives Green energy $O(\sqrt{\varepsilon}(1+|\log\varepsilon|))$. This is stronger than optimizing the source's displayed single-cutoff estimate. We give a general Frostman-exponent version, a flat-boundary calculation, and a conforming primal/flux certificate that separates the computational, boundary-strip, and probabilistic covariance errors. An additional conditional-kernel stability estimate retains exponential moments; small total variation alone cannot control covariance. These are local analytic deductions using classical potential theory and equilibrated flux methods. The stationary tangent law, its positive stiffness, uniform field response, survival reweighting, and stopped-curve identification remain source inputs, not consequences of a finite certificate. Contribution assessment: C3/5, a sharper source-specific boundary-energy estimate and conditional covariance error budget, not an independent proof of the GFF or SLE scaling limits.

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1 The Real Model and Its Normalizations

Let Λ have basis $e_1 = (1, 0)$ and $e_2 = (1/2, \sqrt{3}/2)$, with steps $\pm e_1, \pm e_2, \pm(e_1 - e_2)$. Its vertex density is $v = 2/\sqrt{3}$. On a finite simply connected triangular polygon of mesh n^{-1}, prescribe $+\lambda$ and $-\lambda$ on the two boundary arcs, with $0 < \lambda \le 1/2$. The two marks are boundary-edge midpoints, so no vertex receives two prescriptions. The probability law is normalized Lebesgue measure in the free coordinates of

$$\mathcal{P} = \{h : |h_x - h_y| \le 1 \ (x \sim y),\ h|_{A_+} = \lambda,\ h|_{A_-} = -\lambda\}.$$

This body is bounded, and the zero assignment to free vertices is a strict feasible point for all inequalities involving a free vertex. Fixed–fixed inequalities need only be feasible. Signs of free heights are almost surely nonzero. Affine interpolation gives one chord and possibly loops; the ordered dual sign strand is within $O(n^{-1})$ of the affine chord. Crucially, the exploration filtration records the dual signs, not the real height ratios implicit in affine crossing positions.

We use the source convention

$$X_n(f) = \frac{1}{vn^2} \sum_x f(x/n)(h_x - \mathbb{E}h_x), \qquad G_V = (-\Delta_V)^{-1}, \qquad \sigma^2 = (vA)^{-1}, \quad \lambda_*^2 = \frac{\pi}{8vA}. \tag{1}$$

Here A is the implicit stationary tangent-flux coefficient, not a fitted lattice conductance. On already rescaled coordinates the test is $f(x)$ instead of $f(x/n)$. There is no additional logarithmic normalization of this distributional field, and no assertion of pointwise tightness.

The source claims a GFF limit and, at λ_*, a chordal SLE_4 limit in uniform curve distance modulo increasing reparametrization [1]. We analyze selected mechanisms and prove the results below; we do not certify that complete claim. The reflected bath has generator $\Delta + b \cdot \nabla$ in the free height coordinates and Brownian covariance $2I$. Thus a tilt $\exp(tX_n(f))$ adds drift $tf/(vn^2)$, not tf/n^2. On a free cluster of size H, its orthonormal constant coordinate has Brownian variance parameter two, while its arithmetic mean has variance parameter $2/H$.

1.1 Current public context

Schramm's Problem 2.3 asks specifically about this real hard-constraint model; Problem 2.2 concerns a different integer model [2]. Miller's Theorem 1.1 covers symmetric C^2 potentials with $0 < a \le V'' \le A$ and Lipschitz V'', on the stated square-lattice domain approximations [3]. A flat potential on $[-1, 1]$ that is infinite outside is not in that class. The August 2026 Buchholz–Cotar–Schweiger preprint has a substantially broader nonconvex setting, but its stated assumptions still require positive linear growth of V' and controlled second derivatives [4]. Its abstract and metadata were refreshed, not its full proof. Neither result is silently extended to the present hard constraint. The assigned September 25 manuscript is treated as a publisher claim with an authenticated captured text, not as an independently established consensus result.

2 Source Proof Structure and Retained Obligations

The following map separates the actual model from the continuum inverse used in our certificates.

Slack and moments. The source first uses the zero-pinned controlled-gradient input, triangular facet exposure, log-concave filling volumes, and sparse-hole extensions to prove exponential moments of macroscopic averages. The compressed triangular ray has distance $\max(k, \ell)$, not $k + \ell$; the source expressly excludes a near-tie at its first edge. That valid correction is retained, not counted as a new result here. Bounded pin comparison transfers long-distance slack but does not by itself transfer microscopic strictness. The separate strict-filling argument uses fixed pins in $[-1/2, 1/2]$ and sign intervals containing zero. Arbitrary revealed values in $[-1, 1]$ do not satisfy the same quenched premise.

Reflected comparisons. The common-noise comparison produces weighted edge energy and, at common one-sided walls, weighted squared *values*:

$$\mathcal{E}_n = \sum_{e \in G_n} w_e |\nabla z_e|^2 + \sum_{x \in S_n^{\mathrm{g}}} w_x |z_x|^2. \tag{2}$$

The source bounds its expectation and constructs filled differences with squared $W^{1,p}$ bounds for $1 < p < 2$. Reciprocal-weight spatial moments have deterministic bounds outside negligible events. Merely finite random moment constants could not be multiplied by energy controlled only in L^1. The original–filled error is proved to vanish in L^1 through improved defect parameters; a fixed hole construction is retained throughout.

Adapted exact-value queries, followed by synchronous rejoining of the conditional Gibbs laws, preserve each *annealed* marginal. The proof does not give a uniform estimate for every realized transcript. Shrinking two-sided wall intervals can have strict-edge estimates without the uniform full-dimensional forcing needed in the comparison theorem.

The stationary cell problem. On an isolated regular contact episode the tangent update is exactly

$$P_e = I - \frac{a_e a_e^{\mathsf{T}}}{|a_e|^2}, \qquad J = P_{e_k} \cdots P_{e_1}.$$

The factors average incident coordinates, but their product need not be symmetric. Order is retained. A permutation would preserve squared norm without providing the required contact dissipation. The source uses a stationary bulk environment including the Brownian drivers, controlled cocycles with affine volume growth, a Huber-cutoff zero-slope uniqueness argument, and forward-past contact recurrence. For such an attained slope q its flux is

$$j(q) = \mathbb{E}\left[-\frac{1}{T|C_0|} \sum_{y \in C_0} (y - \overline{C}_0)(Jw - w)_y \right]. \tag{3}$$

Uniqueness, rotations and reflections make this Aq once a nonzero slope is attained. A positive-source response with true zero exterior then establishes $0 < A < \infty$. Finite contact products alone do not establish any of those infinite-law statements.

Traces and conditional covariance. The wall term in (2), transported Frostman measures, and $p > 3/2$ identify zero one-half Frostman trace on the limiting observed set. The source first proves equality with the variational Green solution, and only then concludes H_0^1 membership; it does not promote a $W^{1,p}$ trace directly to an energy trace. Observed limits may have positive area, on which the response must also vanish. Uniform small-tilt response and exponential moments give

$$\kappa_n(\mathcal{C}) := \sup_s \sup_{f,g \in \mathcal{C}} \left| \mathrm{Cov}_{P_{n,s}}(X_n(f), X_n(g)) - \sigma^2 \langle f, G_{Q \setminus K_{n,s}} g \rangle \right| \longrightarrow 0 \tag{4}$$

for each fixed admissible compact smooth-test family. This is an imported probabilistic approximation error in our quantitative certificates. No rate for κ_n is established here.

Geometry and identification. Feasible shifts average the prescription on near-contact clusters and use a tangent-vector-field determinant identity. They transfer an event of positive probability, not create the initial event. Barrier, hookup, separation, survival and nonreturn estimates then support harmonic landing records and height-gap extraction. The stopped harmonic observable is an exact conditional-mean martingale only for a deterministic test fixed before stopping. The proof uses survival reweighting, not absolute continuity alone. Forward and reverse driving convergence, followed by a strong-path criterion and uniform boundary-map convergence, remain distinct steps. The middle barrier/separation and height-gap comparison proofs not literally read in this report remain explicitly imported in Section 7.

3 A Two-Endpoint Boundary Green Estimate

Fix a bounded square $Q \subset \mathbb{R}^2$. Let $K \subset \overline{Q}$ be compact, contain ∂Q, and satisfy the source's continuum thickness condition: for every $x \in K$ and $0 < r < r_0$, there is a continuum

$$x \in L \subset K \cap \overline{B}(x, 2r), \qquad \operatorname{diam} L \geq r. \tag{5}$$

Put $V = Q \setminus K$. The Green operator is the variational Dirichlet inverse on $H_0^1(V)$, extended by zero on K. It is not a free-space fundamental solution. Constants below depend on Q, r_0, the thickness constants, and specified Frostman constants, not on the fine shape or the number of components of V.

We use two classical potential-theoretic facts also used by the source. Disk domination gives

$$0 \leq G_V(x, y) \leq C(1 + \log^+(1/|x - y|)). \tag{6}$$

Planar Beurling and (5) give the escape estimate

$$\mathbb{P}_x\{\tau_{\partial B(x,r)} < \tau_V\} \leq C \min\{1, \sqrt{\operatorname{dist}(x, K)/r}\}, \qquad 0 < r < cr_0. \tag{7}$$

Here Brownian exit probabilities are independent of a constant change of clock. The Green kernel remains normalized by $-\Delta$, as in (1). One obtains (7) by centering at a nearest point of K, using a crossing continuum there, and comparing the two radii by fixed factors. These are potential-theory inputs, not new probabilistic estimates for the height field.

Lemma 3.1 (Two killed endpoint laws)**.** *Write $\delta_x = \operatorname{dist}(x, K)$ and $d = |x-y|$. If $d > C_0 \max(\delta_x, \delta_y)$, then*

$$G_V(x, y) \leq C \frac{\sqrt{\delta_x \delta_y}}{d} (1 + \log^+(1/d)). \tag{8}$$

In particular, for $\delta_x, \delta_y \leq L\varepsilon$ and $d > C_L\varepsilon$, the numerator can be replaced by $L\varepsilon$.

Proof. First suppose d is smaller than a fixed multiple of r_0. Use disjoint balls about x, y of radius $d/8$, and let ω_x, ω_y be their subprobability exit laws before killing on K. Since the pole is outside the respective ball, the strong Markov property for the Green kernel, followed by its symmetry, gives

$$G_V(x, y) = \iint G_V(u, w)\, \omega_x(\mathrm{d}u) \omega_y(\mathrm{d}w). \tag{9}$$

This is two successive harmonic representations. It assumes no independence between two events in one Brownian trajectory. The measures are determined separately by their starting points. Boundary killing contributes zero. The representation can first be made on regular interior exhaustions and then passed by monotonicity. For its remaining points $|u - w| \geq 3d/4$, so (6) bounds the integrand, while (7) bounds the two total masses by $C\sqrt{\delta_x/d}$ and $C\sqrt{\delta_y/d}$. This proves the formula. For larger d, use two balls of fixed radius cr_0 instead. Their Green separation is bounded below; the two probabilities give $C\sqrt{\delta_x\delta_y}$, which implies the stated bound because d is bounded by $\operatorname{diam} Q$. □

Theorem 3.2 (Frostman boundary-strip energy)**.** *Let $0 < \alpha \leq 1$, $M, L < \infty$, and $0 < \varepsilon < \varepsilon_0$. Suppose μ is positive, $\mu(V) \leq 1$,*

$$\mu(B(x,r)) \leq Mr^{\alpha} \quad (0 < r \leq 1), \qquad \operatorname{supp}\mu \subset \{x \in V : \operatorname{dist}(x,K) \leq L\varepsilon\}.$$

Then, with $\ell_\varepsilon = 1 + |\log\varepsilon|$,

$$\mathcal{E}_V(\mu) := \langle \mu, G_V\mu\rangle \leq \begin{cases} C_{\alpha,M,L}\varepsilon^{\alpha}\ell_\varepsilon, & 0 < \alpha < 1, \\ C_{M,L}\varepsilon\ell_\varepsilon^2, & \alpha = 1. \end{cases} \tag{10}$$

The same upper bound holds for signed μ whose total variation satisfies these conditions, for every unnormalized restriction, and after replacing V by any smaller open set.

Proof. Choose $a = C_L\varepsilon$ with the constant in Lemma 3.1, and decrease ε_0 so $a < 1/4$. For a fixed x, dyadic summation and the Frostman bound give

$$\int_{|x-y|\leq a} (1 + \log^+(1/|x-y|))\,\mu(\mathrm{d}y) \leq C_{\alpha,M}a^{\alpha}(1 + |\log a|).$$

For example, partition into $2^{-j-1}a < |x-y| \leq 2^{-j}a$; the resulting sum is bounded by

$$CMa^{\alpha}\sum_{j\geq 0} 2^{-\alpha j}(1 + |\log a| + j).$$

There are no atoms, so the diagonal contributes no mass.

On each outer annulus $r < |x-y| \leq 2r$, with $r = 2^j a < 1$, Lemma 3.1 gives contribution at most

$$CM\varepsilon r^{\alpha-1}(1 + \log(1/r)).$$

For $\alpha < 1$, its sum is at most $C_{\alpha,M}\varepsilon a^{\alpha-1}(1 + |\log a|)$ by the geometric series with ratio $2^{\alpha-1} < 1$. For $\alpha = 1$ there are $O(\ell_\varepsilon)$ annuli, with sum of their logarithmic factors $O(\ell_\varepsilon^2)$. The remaining distances bounded below by a constant contribute $O(\varepsilon)$. Integrate over x, using mass at most one, and combine the bounds.

For a signed measure, positivity of the kernel bounds its absolute quadratic integral by the energy of its total variation. The integrability just proved legitimizes that assertion. For domain restriction, the variational formula and $H_0^1(V') \subset H_0^1(V)$ imply $\mathcal{E}_{V'}(\mu) \leq \mathcal{E}_V(\mu)$, including signed finite-energy functionals. Approximation by smooth measures gives the same conclusion at this level of generality. Restriction preserves the mass and ball bounds; renormalizing a small restriction does not. □

3.1 The actual increment over the source

The source's proof of boundary variance displays, at a cutoff ρ,

$$C\{\rho^{1/2} + (\varepsilon/\rho)^{1/2}\}(1 + |\log \rho|).$$

It takes the two limits in order. Even optimizing that display by $\rho = \sqrt{\varepsilon}$ gives only $O(\varepsilon^{1/4}\ell_\varepsilon)$. Our conclusion at $\alpha = 1/2$ is

$$\mathcal{E}_V(\mu) \le C\sqrt{\varepsilon}\,\ell_\varepsilon. \tag{11}$$

The gain uses both killed endpoint laws and retains separation on each annulus instead of a single split. It is not a new conformal invariance theorem or a claim that the logarithm is sharp. The constants still require certified geometric thickness and the Beurling input. The improvement concerns the continuum covariance term, not the possibly slower original–filled or homogenization errors.

Example 3.3 (A flat boundary, computed rather than simulated)**.** In the upper half-plane let μ_ε be uniform arclength on $\{x + i\varepsilon : 0 \le x \le 1\}$. This has a uniform 1-Frostman bound. The $(-\Delta)^{-1}$ kernel is $G_{\mathbb{H}}(z, w) = (2\pi)^{-1} \log|(z - \overline{w})/(z - w)|$. With $a = 2\varepsilon$,

$$\mathcal{E}_{\mathbb{H}}(\mu_\varepsilon) = \frac{1}{2\pi}\int_0^1 (1 - r)\log(1 + a^2/r^2)\,\mathrm{d}r \tag{12}$$

$$= \frac{1}{2\pi}\left[\frac{1 - a^2}{2}\log(1 + a^2) + \frac{a^2}{2}\log a^2 + 2a\arctan(1/a)\right]. \tag{13}$$

Integrate the logarithm directly, and for its r-weighted integral substitute $s = r^2$. This proves the formula without quadrature. Differentiation in $a > 0$, followed by integration from zero, gives $\int_0^\infty \log(1 + a^2/r^2)\,\mathrm{d}r = \pi a$. Thus $0 < \mathcal{E}_{\mathbb{H}}(\mu_\varepsilon) \le \varepsilon$, and expansion of (13) gives $\mathcal{E}_{\mathbb{H}}(\mu_\varepsilon) = \varepsilon + O(\varepsilon^2(1 + |\log \varepsilon|))$. The half-plane example is a separate exactly soluble Green problem, not a bounded-domain application of the theorem. Bounded subdomains inherit its upper bound when the segment is contained in them. It shows that the logarithms in our general estimate need not be optimal for a smooth boundary. No value of the source coefficient A has been computed.

4 From Boundary Energy to a Covariance Certificate

This section supplies a finite-computation interface, not a procedure for obtaining the source's infinite-volume coefficient. Let V be any open subset of Q, and equip $H_0^1(V)$ with norm $\|\nabla u\|_2$. Poincare on Q, followed by zero extension, gives a coercive norm even if V is disconnected or has an irregular boundary. Write $H^{-1}(V)$ for its full dual. For $f \in H^{-1}(V)$, the weak solution $u = G_V f$ satisfies

$$\int_V \nabla u \cdot \nabla w = \langle f, w\rangle \quad (w \in H_0^1(V)), \qquad \mathcal{E}_V(f) = \|\nabla u\|_2^2.$$

The original Ram elliptic proof explicitly retains the boundary lifting, operator domain and source measure [5]. Its inverse is a weak Dirichlet inverse before any pointwise kernel is asserted. We use that distinction here: rough height samples are not Taylor-expanded.

Proposition 4.1 (Conforming energy enclosure)**.** *Let $f_0 \in L^2(V)$, $w \in H_0^1(V)$, and $\tau \in H(\mathrm{div}; V)$. Suppose a certified bound*

$$\|f_0 - \mathrm{div}\,\tau\|_{H^{-1}(V)} \le \rho$$

is available. Define

$$L = 2\langle f_0, w\rangle - \|\nabla w\|_2^2, \qquad U = (\|\tau\|_2 + \rho)^2. \tag{14}$$

Then $\max(0, L) \le \mathcal{E}_V(f_0) \le U$. *If* $\operatorname{div}\tau = f_0$,

$$U - L = \|\tau + \nabla w\|_2^2 = \|\nabla(G_V f_0 - w)\|_2^2 + \|\tau + \nabla G_V f_0\|_2^2. \tag{15}$$

Proof. Testing the exact weak equation against w gives $\mathcal{E}_V(f_0) - L = \|\nabla(G_V f_0 - w)\|_2^2$. For the upper bound put $r = f_0 - \operatorname{div}\tau$ and let $z = G_V r$. Then $\tau - \nabla z$ has divergence f_0 and norm at most $\|\tau\|_2 + \rho$, since $\|\nabla z\|_2 = \|r\|_{H^{-1}}$. For an exactly balanced flux, integration by parts with the zero-trace solution u gives $\|\nabla u\|_2^2 = -\int \tau \cdot \nabla u \le \|\tau\|_2 \|\nabla u\|_2$. This proves the upper bound. Finally $\tau + \nabla u$ is divergence-free. Its inner product with $\nabla(w - u)$ vanishes, which proves (15). □

This is the classical Prager–Synge mechanism, freshly read in the original Ram flux proof [6], not a new hypercircle theorem. It requires actual conforming functions and actual divergence balance, including interelement normal flux. With a source residual, a norm over a finite test space is a *lower* bound for the full dual norm, not the needed upper certificate. An available Poincare constant permits $\rho = C_P\|r\|_2$, though this may be less sharp. In nonorthogonal moment coordinates the mass and stiffness forms must be transported along with the coefficients [7]. Positive finite Gram matrices do not remove omitted-space errors.

Theorem 4.2 (Interior computation plus an observed boundary strip)**.** *Suppose* $f = f_0 + f_b$, *where* $f_0 \in L^2(V)$ *and the signed measure* f_b *has total variation satisfying Theorem 3.2. Let* B *denote its certified upper energy bound and let* L, U *be as in Proposition 4.1. Then*

$$E_- := \left(\max\{0, \sqrt{\max(0, L)} - \sqrt{B}\}\right)^2 \;\le\; \mathcal{E}_V(f) \;\le\; E_+ := (\sqrt{U} + \sqrt{B})^2. \tag{16}$$

For an admissible deterministic source conditioning s, *suppose the field variance comparison for this* fixed *test has certified error at most* κ:

$$|\operatorname{Var}_{P_{n,s}} X_n(f) - \sigma^2 \mathcal{E}_V(f)| \le \kappa.$$

If additionally $0 < A_- \le A \le A_+ < \infty$, *then*

$$\max\left\{0, \frac{E_-}{vA_+} - \kappa\right\} \le \operatorname{Var}_{P_{n,s}} X_n(f) \le \frac{E_+}{vA_-} + \kappa. \tag{17}$$

Proof. The Green quadratic form is the squared norm in $H^{-1}(V)$. The triangle and reverse triangle inequalities give $(\max\{0, \sqrt{\mathcal{E}_V(f_0)} - \sqrt{B}\})^2 \le \mathcal{E}_V(f) \le (\sqrt{\mathcal{E}_V(f_0)} + \sqrt{B})^2$. Both outer bounds are monotone in the nonnegative interior energy. Insert L, U and then the positive coefficient bounds and κ. □

The three terms have different origins: $U - L$ measures a physical elliptic computation, B measures omission of a boundary strip, and κ measures the probabilistic field-to-Green comparison. This theorem does not supply κ, A_-, or A_+. For the actual finite field, f must be one of the source's admissible smooth tests; using a singular measure in the *Green* estimate does not define a new microscopic height observable. Smooth approximations or smeared landing measures are legitimate when their fixed-scale test family and Frostman bound are checked.

Example 4.3 (An equilateral triangle and its physical metric)**.** Let $\widehat{T} = \{s, t > 0 : s + t < 1\}$ and $(X, Y) = (s + t/2, \sqrt{3}\, t/2)$, mapping it to the unit equilateral triangle T. With $u = st(1 - s - t)$,

$$-\Delta_{X,Y} u = \frac{4}{3}, \qquad u|_{\partial T} = 0, \qquad \mathcal{E}_T(4/3) = \int_T |\nabla u|^2 = \frac{\sqrt{3}}{180}. \tag{18}$$

Indeed $\Delta_{X,Y} = \frac{4}{3}(\partial_s^2 + \partial_t^2 - \partial_s \partial_t)$, and the Jacobian is $\sqrt{3}/2$. The normalized energy form is

$$\frac{1}{\sqrt{3}} \int_T \nabla u \cdot \nabla w = \frac{1}{3} \int_{\widehat{T}} (2u_s w_s - u_s w_t - u_t w_s + 2u_t w_t)\, \mathrm{d}s\, \mathrm{d}t.$$

Since $\int_{\widehat{T}} st(1 - s - t) = 1/120$, the weak load $(1/\sqrt{3}) \int_T (4/3)u$ equals $1/180$. For $w = cu$ and $\tau = -\nabla u$, (14) gives $L = (2c - c^2)\mathcal{E}_T(4/3)$, $U = \mathcal{E}_T(4/3)$ and gap $(1 - c)^2 \mathcal{E}_T(4/3)$. This is an exact polynomial certificate, not a numerical PDE experiment. Replacing the equilateral metric by the Euclidean (s, t) metric would change both its load and energy normalization. The example solves a continuum Poisson problem; it is not an exact finite-volume Lipschitz covariance.

4.1 The source's iterated local variance bound

In the source's ambient-domain localization, for fixed $3/2 < p < 2$ put $\beta_p = 1 - 3/(2p)$. The original/local field comparison gives $\limsup_n \|Z_{n,\eta}\|_2 \le C_p \eta^{\beta_p}$. Combining its variance triangle inequality with Theorem 3.2, at $\alpha = 1/2$, yields the conditional consequence

$$\limsup_{n \to \infty} \sup \operatorname{Var} X_n(\mu_{n,\eta}) \le \left(C\eta^{1/4} \sqrt{1 + |\log \eta|} + C_p \eta^{\beta_p} \right)^2. \tag{19}$$

All original anchoring, thickness, moment and comparison premises are retained. For each fixed η, the source first passes the mesh limit in a compact smooth-test family. We do not assert a uniform error for $\eta = \eta_n$. Since $\beta_p < 1/4$, the second term can dominate. Constants need not remain bounded as $p \uparrow 2$. Thus our improved continuum energy does not automatically improve the leading power in this complete localization bound.

5 Approximate Records Must Control Their Actual Laws

For a genuine sign exploration, each next query and stopping decision is measurable in the previously observed signs and independent randomness. Disintegration leaves uniform Lebesgue density on the recorded sign polytope. In the notation of the source, $\mathrm{Law}(h \mid S = s) = P_{n,s}$. Neither an arbitrary field-dependent choice of a string nor an affine crossing position has this property automatically. We now give a controlled weakening of this equality for the covariance interface only.

Lemma 5.1 (Tail-aware covariance stability)**.** *Let P, Q be probability laws on the same measurable space and let F, G be real observables. For some $a > 0, M \ge 1$, assume, under each law and for each observable,*

$$\mathbb{E} e^{a|F|} \le M, \qquad \mathbb{E} e^{a|G|} \le M.$$

Set $\delta = \mathrm{TV}(P, Q) = \sup_E |P(E) - Q(E)|$ and $\phi(\delta) = \delta(1 + \log(1/\delta))^2$ for $\delta > 0$, with $\phi(0) = 0$. Then

$$|\operatorname{Cov}_P(F, G) - \operatorname{Cov}_Q(F, G)| \le \frac{128(M + M^2)}{a^2} \phi(\delta). \tag{20}$$

Proof. For a single observable Y with the exponential bound and $T \geq 0$, layer cake gives

$$\mathbb{E}[|Y|\mathbf{1}_{|Y|>T}] \leq M(T + a^{-1})e^{-aT}, \qquad \mathbb{E}[Y^2\mathbf{1}_{|Y|>T}] \leq M(T^2 + 2T/a + 2/a^2)e^{-aT}.$$

Clip Y to $[-T, T]$. The mean difference of the clipped variables is at most $2T\delta$, and the second-moment difference is at most $2T^2\delta$. Add the preceding tails under both laws. Also $|\mathbb{E}_P Y|, |\mathbb{E}_Q Y| \leq M/a$, so the difference of squared means is at most $2M/a$ times the mean difference. For $\delta > 0$ take $T = a^{-1}\log(1/\delta)$. The resulting variance bound is at most $32(M + M^2)a^{-2}\phi(\delta)$; this intentionally loose constant includes $\delta = 1$. For $\delta = 0$ equality of the laws settles the claim.

Apply the variance bound to $F + G$ and $F - G$, with exponent $a/2$. Cauchy–Schwarz gives their exponential moment at most M. Polarization and a factor $1/4$ then give a bound no larger than the displayed constant. □

Corollary 5.2 (Conditional-record error budget)**.** *Let the actual conditional field kernel be $Q_{n,s}$, with record law $\lambda_n(\mathrm{d}s)$, and let $P_{n,s}$ be the source reference kernels. Suppose the exponential hypotheses of Lemma 5.1 hold uniformly in s for both kernels and the chosen pair of observables. If $0 \leq \delta_n \leq e^{-1}$ and $\int \mathrm{TV}(Q_{n,s}, P_{n,s})\lambda_n(\mathrm{d}s) \leq \delta_n$, then the averaged absolute conditional covariance error is at most $128(M + M^2)a^{-2}\phi(\delta_n)$. For a fixed admissible test family, add this error to the source's κ_n in (4).*

Proof. Apply the lemma conditionally. The function ϕ is concave on $(0, 1]$, since $\phi''(d) = 2\log(d)/d \leq 0$, and continuous at zero. Jensen bounds the average by $\phi(\overline{\delta}_n)$. If merely $\overline{\delta}_n \leq \delta_n$, use $\delta_n \leq e^{-1}$, where ϕ is increasing; for larger upper bounds replace the right side by $\max_{0\leq d\leq\delta_n} \phi(d)$. In particular the displayed asymptotic error holds once $\delta_n \downarrow 0$. If δ_n denotes the actual average rather than an upper bound, Jensen holds for every $\delta_n \leq 1$. □

The small-δ_n qualification is part of this corollary; it is not a claim that ϕ is increasing on all of $[0, 1]$. Means must be centered consistently: one can use the source-centered $X_n(f)$ under both kernels, since covariance is invariant under the same deterministic shift. The moment premise is then required under the actual kernel too. A source moment bound alone does not provide it. The conclusion is a covariance comparison, not an approximate martingale theorem or a replacement for survival.

Example 5.3 (Two elementary obstructions)**.** With one free vertex surrounded by zero pins, the real Lipschitz law is uniform on $[-1, 1]$. A record revealing $|h| > b$, $0 < b < 1$, has conditional variance $(1 + b + b^2)/3$ rather than $1/3$. Relabeling its realized geometry as the original no-record geometry does not preserve the conditional kernel.

For the tail issue, let $P = \delta_0$ and $Q_N = (1 - N^{-2})\delta_0 + N^{-2}\delta_N$. Then $\mathrm{TV}(P, Q_N) = N^{-2}$ but $\mathrm{Var}_{Q_N}(Y) = 1 - N^{-2}$. The exponential-moment premise fails. These are diagnostic probability examples, not examples of the claimed scaling limit failing. The source already warns against arbitrary field-dependent selection; the added content here is a quantitative, tail-controlled covariance tolerance.

6 What the Certificates Do Not Identify

The clock, measure and boundary distinctions survive every local calculation. The source's exact product $P_{23}P_{12}$ on three coordinates sends $(1, 0, 0)$ to $(1/2, 1/4, 1/4)$, whereas $P_{12}P_{23}$ sends it to $(1/2, 1/2, 0)$. Their square losses are respectively $5/8$ and $1/2$. An unordered list of contacts cannot determine (3). Nor does a finite list of such products identify the stationary environment, all attained slopes, or the expectation defining A.

The source normalization of the interface follows only after its field and stopped-curve inputs. For $Z_t(z) = g_t(z) - 2B_t$, the harmonic angle with height gap l has bracket density

$$\frac{16l^2}{\pi^2} \operatorname{Im}(1/Z_t(z)) \operatorname{Im}(1/Z_t(w)),$$

while the loss of the $(-\Delta)^{-1}$ Green kernel is $\frac{2}{\pi} \operatorname{Im}(1/Z_t(z)) \operatorname{Im}(1/Z_t(w))$. The law of total variance, after integrating against protected smooth tests and using the actual conditional kernel, yields $l^2 = \pi\sigma^2/8$. We retain this source calculation; we have not proved its joint convergence premises. In particular, a small flux residual cannot prove the height gap or an SLE driving law. A finite covariance matrix cannot certify a local-set property.

Thickness also cannot disappear from the analytic result: $\log(1/|x|)$ on the punctured disk is in $W^{1,p}$ for $p < 2$, harmonic away from the point and zero on the outer circle. A singleton does not have the required continuum thickness. The source explicitly includes this obstruction. Our rate estimates therefore do not apply uniformly to arbitrarily many microscopic isolated observed islands.

6.1 Advantages, costs and further work

The improved strip bound is uniform in the complicated observed geometry once its thickness constants are fixed. It applies to unnormalized restrictions and to additional observations by domain monotonicity. The certificate (16) can use any conforming approximation and honest equilibrated flux, including moment coordinates with their physical metric. The record error in Corollary 5.2 is explicit and does not require a pathwise coupling of the height samples.

There are real costs. A thin restriction renormalized to mass one can lose its Frostman constant. Flux balance and source dual norms are whole-space statements. The constant A remains implicit, and κ_n may be non-effective. The logarithmic factor in (11) is not asserted optimal. Exponential-tail control under an approximate exploration law is a new verification obligation, not a consequence of a small histogram error. No runtime improvement or finite sampler is claimed.

Useful next steps are to remove the logarithm under additional regularity of the observed set, to certify source-specific $\kappa_n(\eta)$ without interchanging the scale limits, and to construct equilibrated fluxes on slit domains with a rigorous boundary approximation error. Quantitative stationary-law approximations for A would require separate mixing and uniform-integrability estimates, not just contact-word arithmetic.

7 Evidence Scope and Contribution

The full paper assembly, foundations, coupling, cell, both trace sections, shift section, figure TeX and README were literally read in bounded displays. In the long geometry section the displayed intervals were 1–312 and 1198–1340; lines 313–1197 remain unread. In the interface section the intervals were 1–178 and 517–960; lines 179–516 remain unread. Thus the whole architecture is mapped, but the middle barrier/separation and height-gap reading proofs are imported rather than certified. The figure code and caption were read; its separately rendered source figure was not.

Fresh original private readings were TR8 lines 81–197, TR250 lines 457–590, and the physical-metric chapter lines 1–210. They provide weak inverse and boundary conventions, equilibrated flux identities and residual limitations, and physical Gram accounting. Other parts of these originals and the broader corpus were not read. Exact pre-read/post-read hashes, displayed chunks and unread complements are in the owned ledger. Public snapshot authentication is provenance, not a proof audit or live-branch claim.

Owned rational checks cover the equilateral metric and primal/flux identities, contact-word order and dissipation, dyadic energy majorants, covariance perturbation bounds and the flat-boundary formula using explicit logarithm/arctangent enclosures. Cases and assertions are recorded separately. They are finite formula checks, not Brownian simulations, source-checker replays or verification of the probabilistic scaling theorem.

Applicability remains L3. Contribution C3 is assigned to the two-ended boundary-energy improvement and its explicit computational/conditional-law covariance interface. The inverse, hypercircle, Beurling and entropy-tail tools are classical and credited. No source error has been established in the scope examined; a stronger local bound is not evidence that the source's weaker bound was invalid. The unread global proofs and all the retained stochastic uniformity requirements preclude a global certification claim.

References

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[5] M. Subbarao, original confidential `TR8-lin-non-lin.tex`, lines 81–197. Elliptic weak domains, boundary liftings, Green equivalence and discrete source measures. Local reading only.

[6] M. Subbarao, original confidential `TR-rev-1-old-tr250.tex`, lines 457–590. Conforming moment spaces, classical hypercircle proof and source residual estimates. Local reading only.

[7] M. Subbarao, original confidential RamCalBooks, volume 21, `ch01d-physical-metrics.tex`, lines 1–210. Physical Gram forms and omitted residual components; included figure not separately viewed.